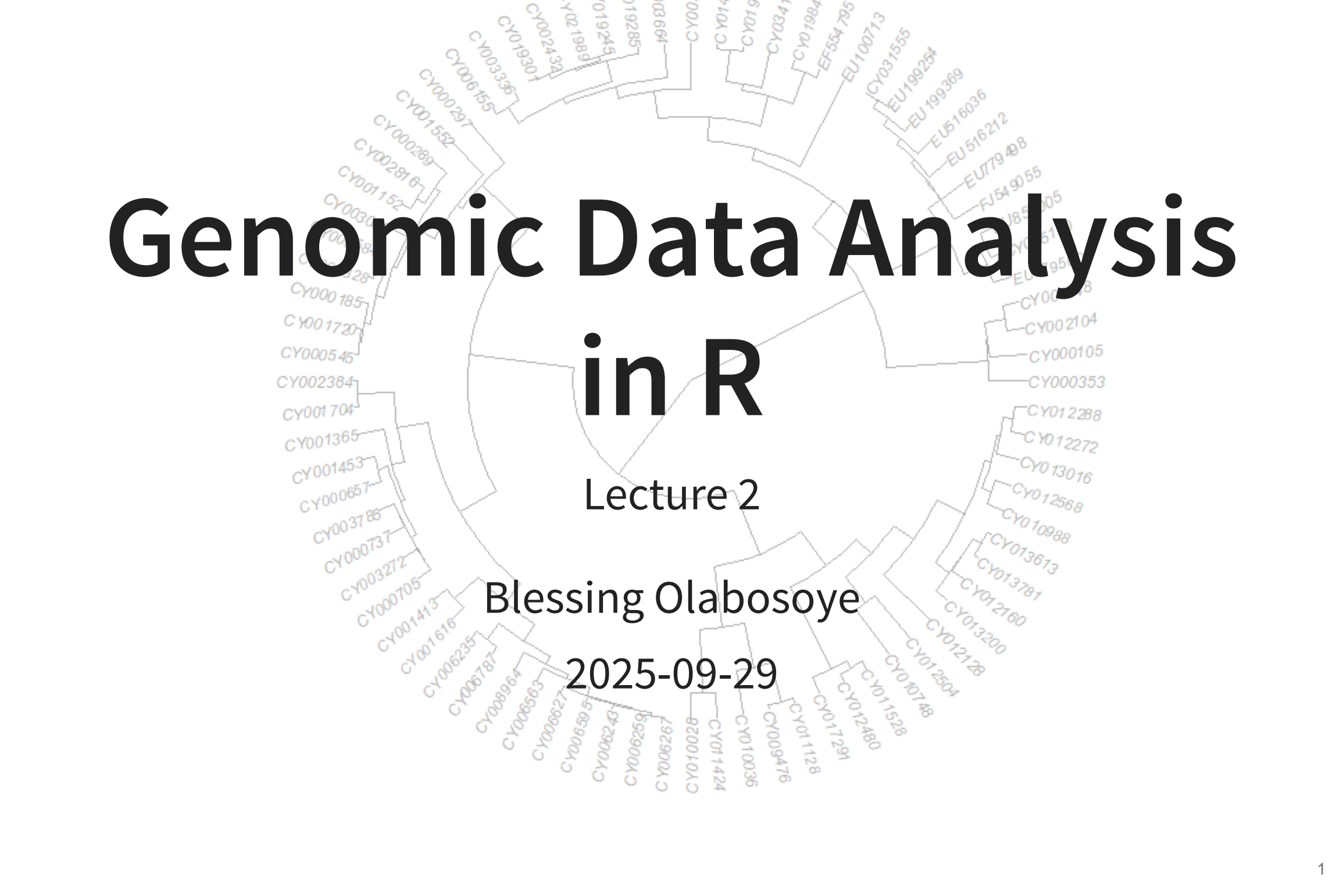


A circular phylogenetic tree (chronoclock) is shown in the background, with numerous accession numbers labeled around its perimeter. The tree is rendered in a light gray color, and the labels are also in light gray, following the curve of the tree's branches.

Genomic Data Analysis in R

Lecture 2

Blessing Olabosoye

2025-09-29

Learning outcomes

- Understand sequence alignment and phylogenetic analysis
- Perform pairwise and multiple sequence alignment for difference data
- Recognize different methods for constructing phylogenetic tree
- Perform phylogenetic analysis and visualize using different graphical representation
- Understand the ethics and privacy of genomics data
- Create a reproducible and publication-ready outputs

Sequence Alignment

Sequence alignment is the process of matching sequences (e.g, DNA, RNA, or protein) to reference sequences in order to identify similarity and to understand functional, structural, and evolutionary consequences. The sequence input is called **query sequence**.

Two categories of alignment:

- Global alignment: Matches the entire length of the query sequence to the reference sequence
- Local alignment: Matches a segment of the entire sequence to the reference sequence

Methods of alignment

- Word matching: Using a short non-overlapping query word, alignment is sort for with the reference sequence. It's known for its computational efficiency
- Dynamic programming: It uses a substitution matrix to assign scores (positive mismatch, negative mismatch, or gap penalty). For protein sequences, gap penalty is given to amino-acid to gap matches between query and reference sequences. For local and global alignment, Smith-Waterman and Needleman-Wunsch algorithm is used respectively.
- Dot-matrix methods: Using a dot-matrix plot, it identifies similarity between sequences with sequences with the highest similarity score in the main diagonal.
- Maximal unique matching

Types of alignment:

- Pairwise alignment: This entails aligning two sequences to obtain similarities and differences between which can provide insights into their evolutionary changes
- Multiple sequence alignment. This is similar to pairwise. Whereas pairwise alignment is for two sequences, multiple alignment can align more than two sequences to observe similarity among sequences and understand evolutionary patterns.

PREDICTED: Drosophila yakuba breast cancer type 2 susceptibility protein homolog (LOC6532063), transcript variant X1, mRNA

Sequence ID: [XM_002092804.4](#) Length: 3658 Number of Matches: 1

Range 1: 6 to 732 [GenBank](#) [Graphics](#)

▼ [Next Match](#) ▲ [Previous Match](#)

Score	Expect	Identities	Gaps	Strand
547 bits(296)	1e-150	601/746(81%)	30/746(4%)	Plus/Plus
Query 21	CAAAGCAACGCTTACCAACACTGCACCCCTTGTTTTGGAAATGCAACCACAAAAGCATTG	80		
Sbjct 6	CAAAGCTACGTTTGCCAACACTG-A-ATCTTGTTTTGGAAATGCAACCACAATTAAATTG	63		
Query 81	GACACTTGCTTACTTCAAATAAA--ACACATTTAAACAATTAGATGACGTGATGGACCAG	138		
Sbjct 64	GCCACTTGTTTACCCCAAATAAAGTACACA-AT-CACAATTAAATGACGTGATGGACCAG	121		
Query 139	AATGGCGCATCGGGAAGTCATCCGAACAGGCTATCCCAAGGAAGAGGCCCATGCGCGC	198		
Sbjct 122	AAAGGCGCTTCGGGAAGTCATTCGACCCGGCTATCCCAAGGAA-A--AG-TCA--C-C-C	173		
Query 199	GAACGTGGCGCCACAGTTTCCGCGGGCGCAAATC-G-G-AGTAACATTATCGACGAAATG	255		
Sbjct 174	--AC--GCGCAGCAG--GCCGCGGGGAAAGTCGGCGAAGTAACATCATCGAGGACATG	226		
Query 256	GCCAAAATATGCGAAGCCGATCGCCAGACTTTGCCATCGCTCGACGGACTCGGGGTCAC	315		
Sbjct 227	GCCAAAATATGCGAAGCCGATCGCCAGACTAGCTCCTTCGCTCAAAGGACTCAAGGTATC	286		
Query 316	GAGCGGCTTGGGTGGACAACAGCGACTTCGTCGCCGTGGAGGATCTTATTTGTCCTAC	375		
Sbjct 287	GAGCGGCTTTCGGCTGACAACAGTGGCTTCGTCGCCGTGGAGTACCTTATTCTGGCCAGC	346		
Query 376	GCAGAGCCACGCCCGAGGACCAGGTCGAGATGATCATGAGCGACTTTTGCTCGTCTCCA	435		
Sbjct 347	GCTGAGCCACGCCCGAGGAGCGGGTTGAGATGATCATGCTGGACTTTTGCTCGTCTCCA	406		
Query 436	ACATACGCAGAGGATGAGGATGAGCCAGCCATGAGTCGGAGCCGTGGTTTCGATTTTCGC	495		
Sbjct 407	ACTTACACAGAGGATGAGGATGAGCCTAGCTATGAGTCGGAGCCGTGGTTTCGATTTTCGC	466		
Query 496	AACAAAAGGATCAGAACCTACAGCCGGAAGAGGGATCCAAAAGCCACAAGGCCGTTCAA	555		
Sbjct 467	AAGAAAAGGATCAAAACTTACGGTCGGAAGAGAGATACAAAAGGTCATATGGTCCTGCGG	526		
Query 556	AACGAGAAGCGTAGAGGTTCTCAGGCCTCTCCGTGCAGAGGGATCTCAATACTTCGTTT	615		
Sbjct 527	AACGAGGAGCCCAATAGTTTCTCGGCCCTCTCTATACAGAGGGATCCCAATAGTTCGTTG	586		

Alignment softwares

- BLAST: identifies similar regions between biological sequences query and databases (reference) sequences.
 - blastn: nucleotide query to nucleotide reference (database)
 - blastp: protein query to protein reference
 - blastx: translated nucleotide query to protein reference
 - tblastn: protein query to translated nucleotide reference
 - tblastx: translated nucleotide query to translated nucleotide reference
- DECIPHER: A R package

Phylogenetic Analysis

Phylogenetic is study of the evolutionary history of relationship or relatedness between organisms using either their physical characteristics or genetic information e.g DNA, RNA, or protein sequences.

- Taxa: any group of an organism
- Clade: a group of organisms consisting of a common ancestor and all its descendants. It is also called monophyletic group.

Phylogenetic tree

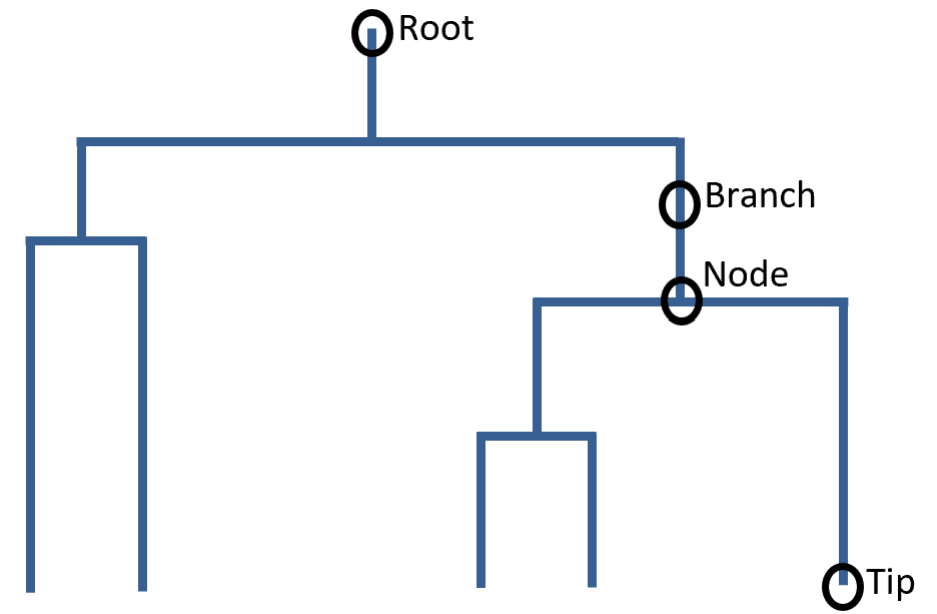
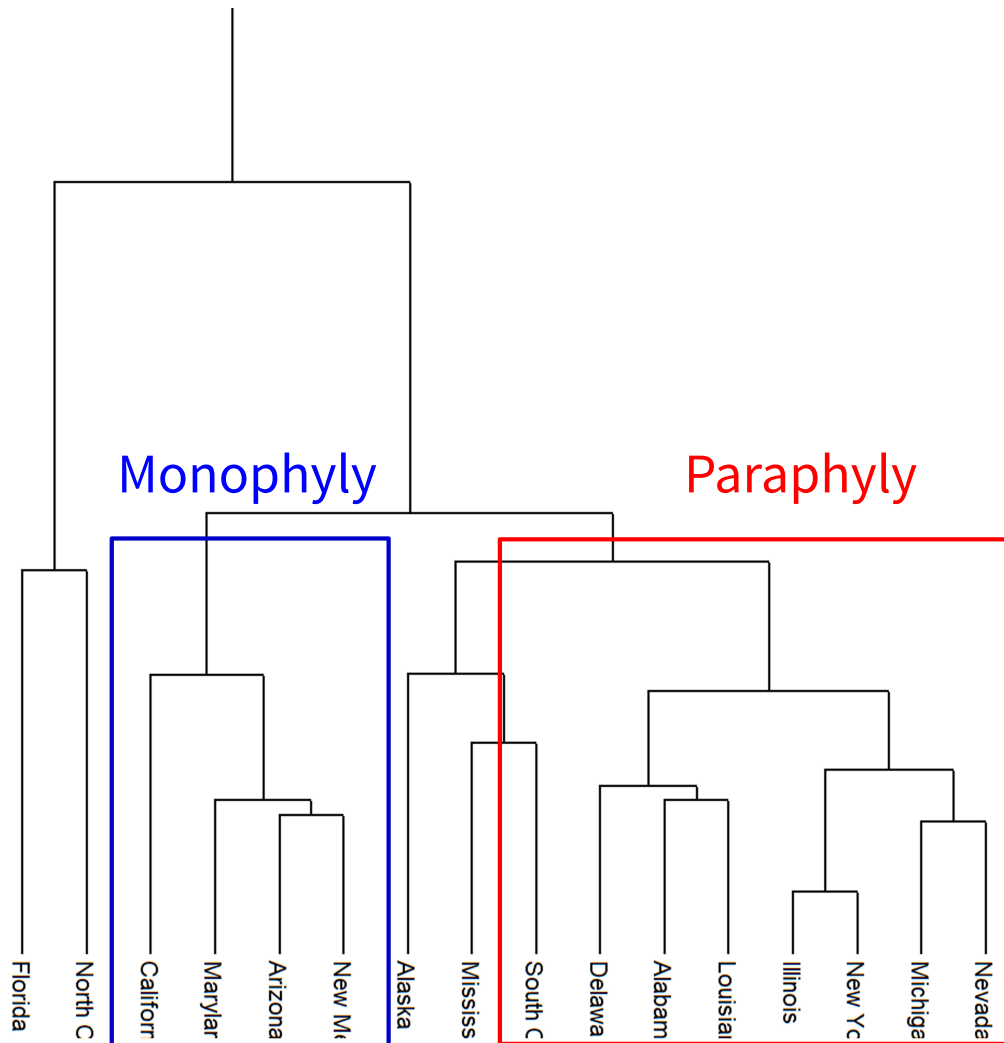
A phylogenetic tree provide a graphical representation of the evolutionary history of a taxa at a specific time based on physical characteristics or genomic information. It aims to infer ancestry patterns between species or represent relationships among genes or proteins or individual organisms.

- The tree can be rooted by having a node (common ancestor) or unrooted with no node but rather networks of interactions.

Terms:

- Monophyletic group is a group consisting of all the descendants of a common ancestor
- Paraphyletic group is a group consisting some (but excludes other) descendants of a common ancestor
- Polyphyletic group is a group consisting of both descendants and non-descendants of a common ancestor

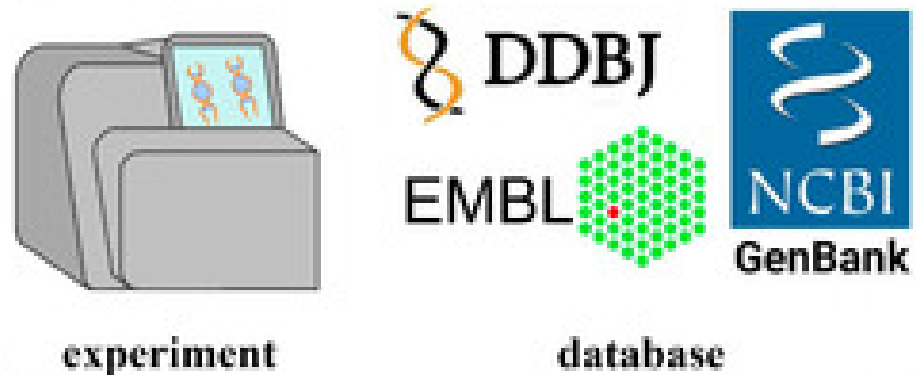
Phylogenetic tree composition



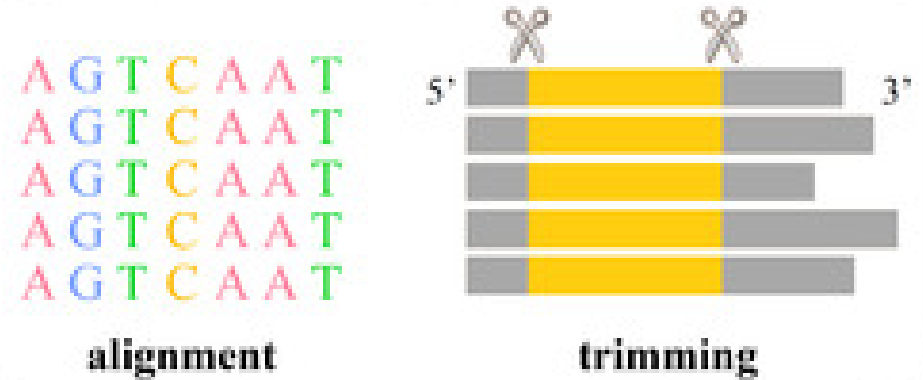
Data source: USArrests

Workflow: Phylogenetic tree construction

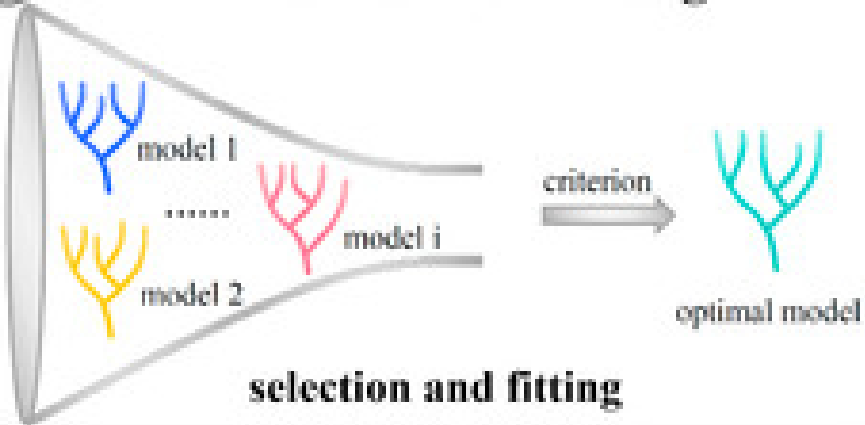
① Sequence Data Collection



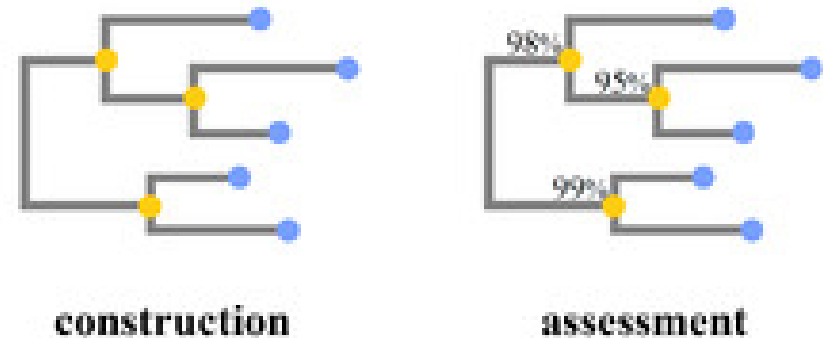
② Sequence Alignment & Trimming



③ Model Selection & Fitting



④ Tree Construction & Assessment



Methods for constructing phylogenetic trees

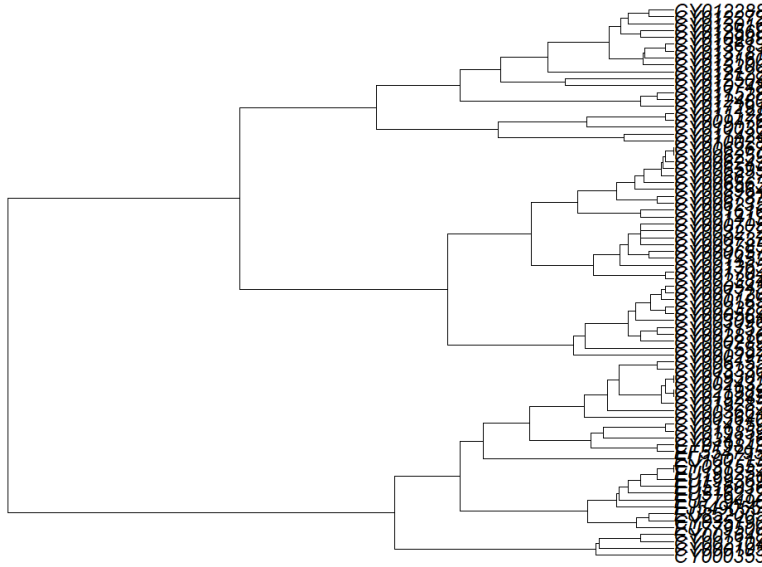
- Distance-based: Given a feature matrix, it converts to a distance matrix to infer evolutionary distances between a given pair of features (e.g, species, DNA, or RNA). The distance can be calculated using Euclidean, Hamming, Manhattan, or Jaccard distance metric. By forming the initial tree and merging trees together, the process continues until a single cluster remains.
 - Neighbor-joining: It uses agglomerative hierarchical clustering to build phylogenetic trees based on minimum evolution criterion. Its tree can be rooted or unrooted dendrogram.
 - Unweighted pair group method with arithmetic mean (UPGMA): It also uses agglomerative hierarchical clustering to build phylogenetic trees. Its tree is based on average evolution distances and it produces a rooted dendrogram.

- Character-based: A large number of hypothetical trees are generated based on a given algorithm and eventually optimal evolutionary tree is inferred. The common algorithms include:
 - Maximum Parsimony: This seeks to infer evolutionary tree by minimizing number of evolutionary changes needed to explain the data. This can be done using Ockham's razor principle of parsimony
 - Maximum likelihood: Given an hypothetical tree, what's the probability that that it comes from an observed data. It requires an evolutionary model and likelihood function of the hypothetical tree given the observed data. The goal is to find the optimum tree.
- Model-based:
 - Bayesian inference: This uses Baye's theorem to identify the optimum hypothetical tree for evolutionary history. $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$

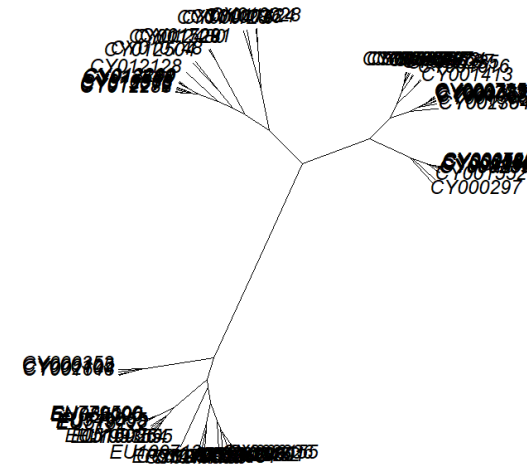
Phylogenetic tree layout

- Rooted tree: Ancestor is emphasized and branch length can represent time
- Unrooted tree: Ancestry is not inferred but only relatedness
- Cladogram tree: Only branching pattern as ancestry is not inferred and branch length does not imply time
- Others layouts: Fan and Radial

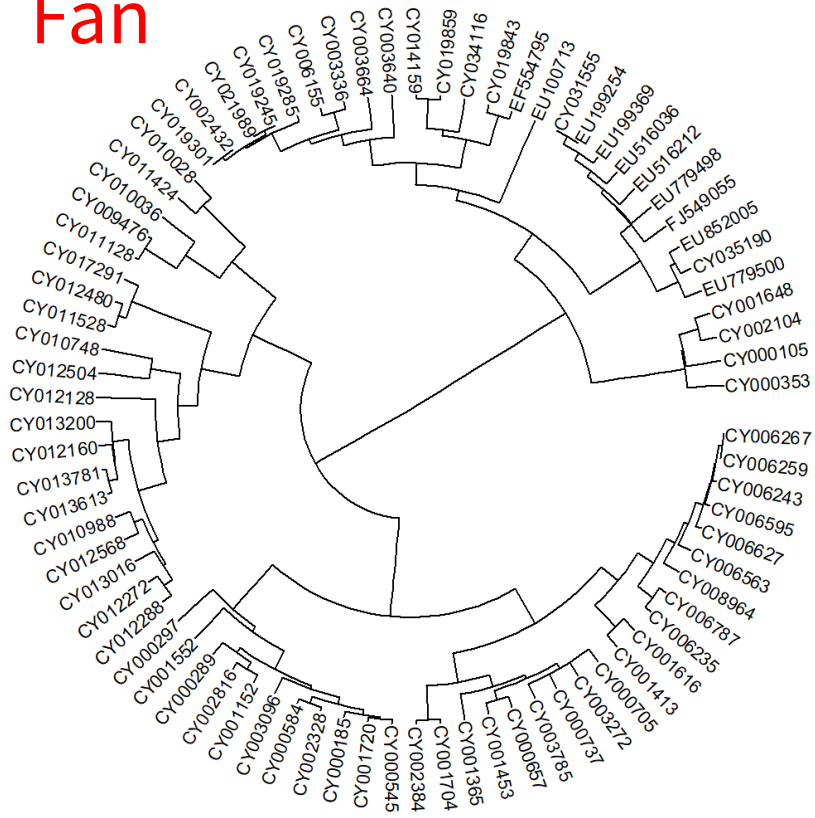
Rooted



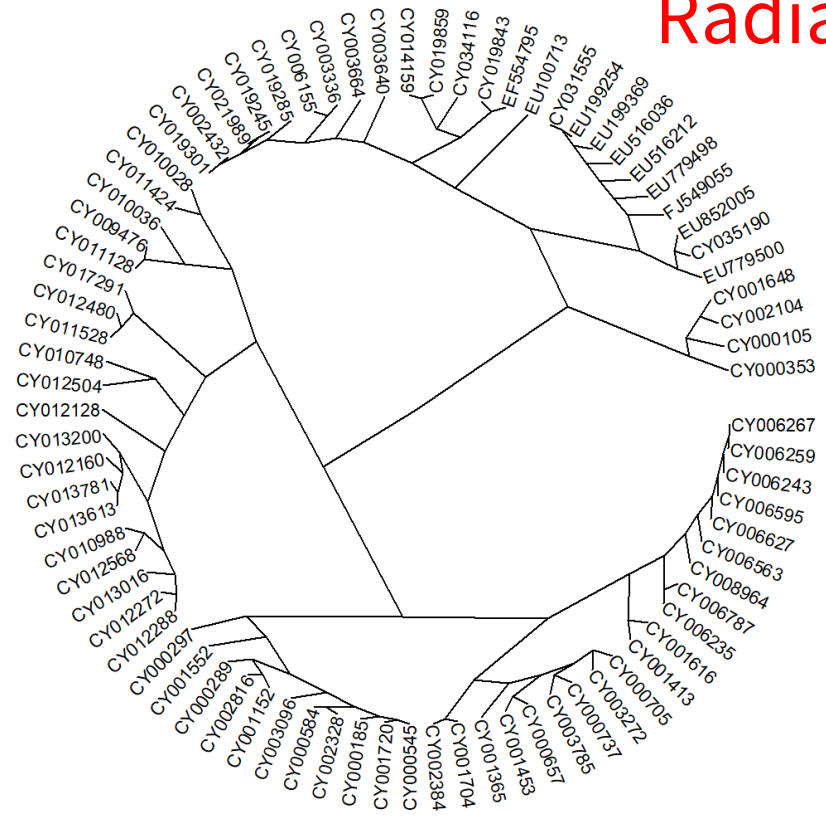
Unrooted



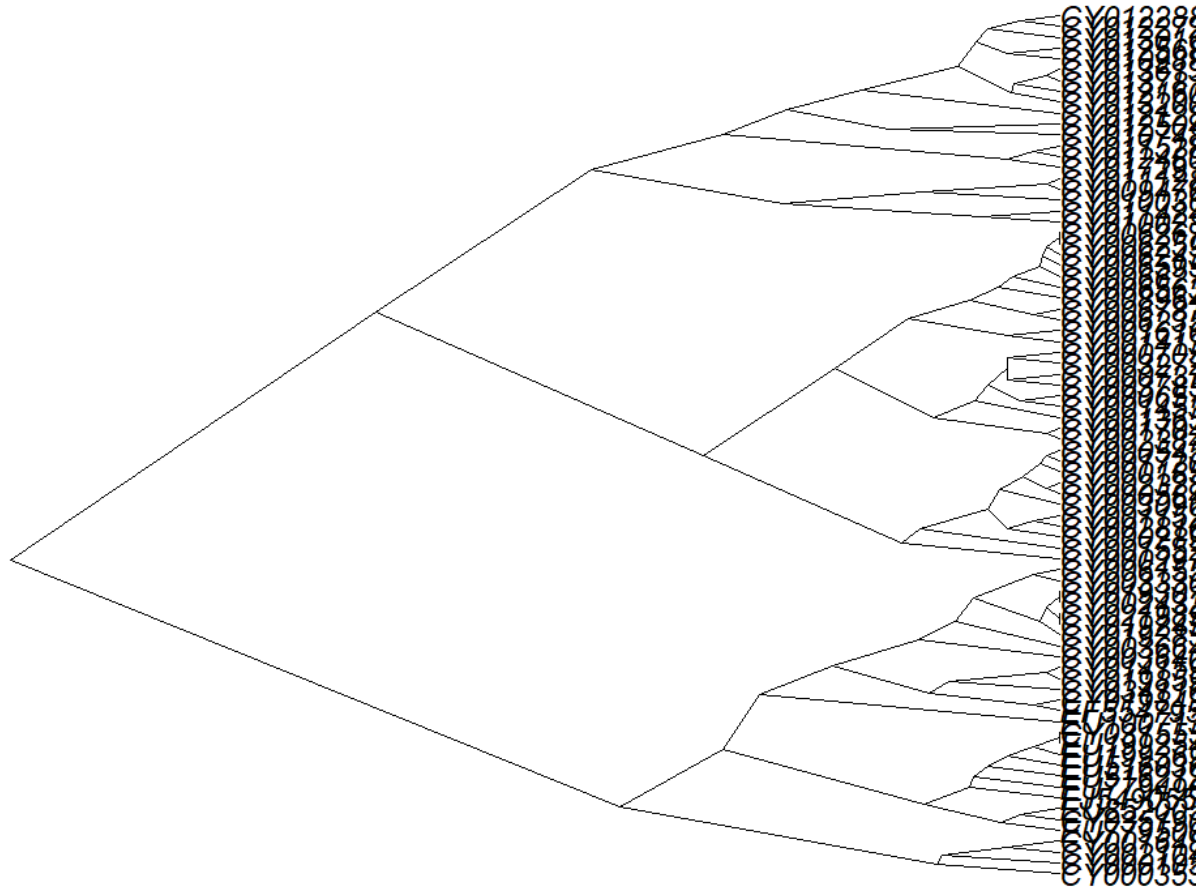
Fan



Radial



Cladogram



Common issues in Phylogenetics:

- What is the best evolutionary models? To address this, consider model selection criteria such as Likelihood-ratio test, Akaike information criterion (AIC), or Bayesian information criterion (BIC)
- Misinterpretation. Caution needs to be taken as the resulting trees are hypothetical and further evaluations are needed

Ethical considerations for genomic data

The ever-increasing production of genomic data has to a large extent contributed and contributing to societal good. However, the potential damage it could cause should be considered carefully.

- Data ownership
- Data privacy
- Consent
- Data sharing
- Government policy
- Population inequality
- Environmental impact

Resources:

- [Molecular Phylogenetics Techniques](#)
- [The Epidemiologist R Handbook](#)
- [adeget](#)